PYTHON DATA UTILITY PACKAGE

Turn raw data into a clearer starting point

Automate common dataset checks before deeper analysis or machine-learning work.

THE PROBLEM

EDA takes repeated effort

- Analysts repeatedly check types, missing values and statistics.
- Outliers and target imbalance can be overlooked.

THE SOLUTION

One repeatable workflow

- Create a formatted Excel EDA report from a pandas DataFrame.
- Run separate checks for ML readiness.

THE BENEFIT

Faster first-pass checks

- Consistent checks across datasets and projects.
- Readable reports for review and hand-off.

HOW IT FITS INTO THE WORKFLOW

1. **Load data** — pandas DataFrame
2. **Generate EDA** — generate_eda_report()
3. **Check readiness** — assess_ml_readiness()
4. **Review outputs** — Excel workbooks

Designed for: **Data analysts • Learners • Project teams • ML practitioners**

A practical pre-analysis helper—not a replacement for domain knowledge or model validation.

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GitHub: EDA Prep Library repository

Two functions. Two complementary views.

Understand the dataset first, then check for common issues before training a model.

1. generate_eda_report()

Dataset profiling → Excel workbook

- Column data types and missing-value counts
- Descriptive statistics and value counts
- Duplicate rows and IQR-based outlier checks
- Optional plots and separate worksheets

2. assess_ml_readiness()

Pre-model checks → readiness report

- Review target suitability for the selected task
- Highlight target-class imbalance in classification
- Surface data-quality and modelling considerations
- Export findings to Excel for review

TECH STACK

Python	Core package
pandas	DataFrames
NumPy	Numerical work
openpyxl	Excel output
Matplotlib / Seaborn	Plots
setuptools	Packaging

WHY IT HELPS

- Repeatable checks reduce manual effort
- Consistent reports improve hand-offs
- Early flags guide the next investigation

IMPORTANT

Readiness results are guidance, not a guarantee. Validate assumptions, leakage, data quality and model performance for each use case.

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Open the EDA Prep Library on GitHub